

Supplementary Material: Calibration and Top-N Ablation Analysis

Overview

This supplementary document provides additional calibration analyses and ablation results. We include combined Top-N performance curves, Expected Calibration Error (ECE), and reliability diagrams across all ensemble sizes. These results show that larger prompt ensembles improve classification performance and Brier-score calibration, while ECE remains better than the single-prompt setting across larger ensembles.

Top-N Performance Comparison

Figure 1 shows the combined accuracy, macro-F1, and Platt-scaled Brier score across all Top-N configurations. Both accuracy and macro-F1 improve with larger ensembles, while the Brier score decreases, indicating better confidence calibration.

Expected Calibration Error (ECE)

Figure 2 reports the Expected Calibration Error across Top-N values. ECE decreases substantially from Top-1 to Top-5 and remains below the Top-1 value at larger ensemble sizes, indicating improved calibration relative to single-prompt inference.

Reliability Diagrams

Figure 3 presents reliability curves for all Top-N settings. Larger ensembles produce curves closer to the diagonal, indicating improved alignment between predicted confidence and empirical accuracy.

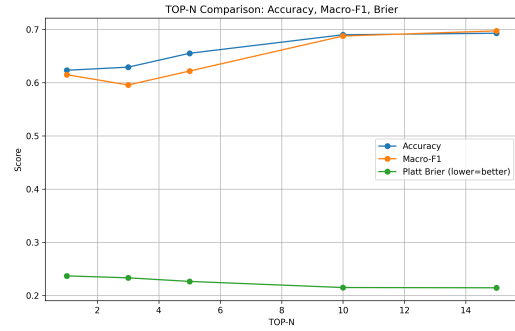


Figure 1: Combined Top-N comparison showing accuracy, macro-F1, and Platt Brier score. Larger ensembles yield improved performance and better calibration.

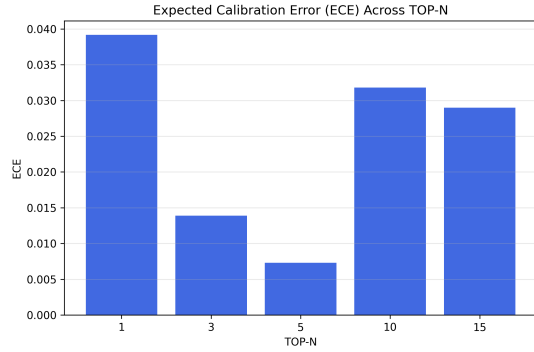


Figure 2: ECE improves from Top-1 to Top-5 and remains lower than the single-prompt setting for larger ensembles.

Table 1: Summary of performance and calibration metrics across Top-N.

Top-N	Accuracy	Macro-F1	Weighted-F1	Platt Brier	ECE
1	0.623	0.615	0.615	0.23696	0.039174
3	0.629	0.596	0.596	0.23330	0.013901
5	0.655	0.622	0.622	0.22656	0.007315
10	0.690	0.687	0.687	0.21508	0.031785
15	0.693	0.697	0.697	0.21460	0.028978

Table 2: Comparison between RemoteCLIP (Top-N=15) and fold-safe RGB-only baselines.

Method	Accuracy	Macro-F1	Weighted-F1
RemoteCLIP (Top-N=15)	0.693	0.697	0.697
GR-NDI	0.359	0.225	0.225
ExG	0.255	0.151	0.151
Turbidity ratio (R/G)	0.359	0.225	0.225

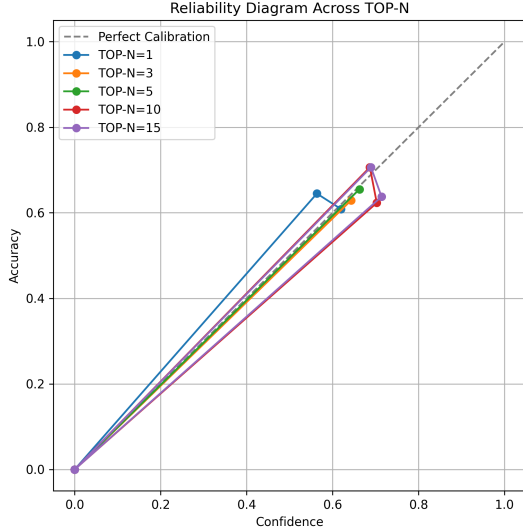


Figure 3: Reliability diagrams across Top-N. Prompt ensembling generally improves confidence calibration relative to single-prompt inference.

Summary of Calibration Metrics

Table 1 summarizes accuracy, macro-F1, Brier score, and ECE for each Top-N configuration.

RGB-only Baseline Comparison

Because our dataset consists of RGB aerial imagery rather than multispectral satellite data, traditional spectral indices such as NDWI or AWEI cannot be computed due to the absence of NIR and SWIR bands. We therefore evaluate three lightweight RGB-only proxy baselines: a green-red normalized difference index (GR-NDI), Excess Green (ExG), and a red-green turbidity ratio. These baselines are not intended as full five-class classifiers; rather, they provide simple heuristic comparisons based on coarse color cues.

For each tile, we compute one scalar score from the valid RGB pixels:

$$I_{\text{GR-NDI}} = \text{mean}\left(\frac{G - R}{G + R + \epsilon}\right),$$

$$I_{\text{ExG}} = \text{mean}(2G - R - B),$$

$$I_{\text{Turb}} = \text{mean}\left(\frac{R}{G + \epsilon}\right),$$

where R , G , and B denote red, green, and blue pixel intensities, and ϵ avoids division by zero.

Each scalar proxy is converted into a binary prediction using a threshold selected only within the

training folds and then applied to the corresponding held-out fold. GR-NDI maps tiles to *deep water* or *turbid water*; ExG maps tiles to *submerged aquatic vegetation* or *rocky substrate*; and the turbidity ratio maps tiles to *turbid water* or *deep water*. Because these proxies are inherently binary, they cannot represent all five hydrological classes simultaneously and are evaluated independently against the full five-class ground truth.

As shown in Table 2, the best RGB proxies, GR-NDI and the turbidity ratio, achieve 0.359 accuracy and 0.225 macro-F1, while ExG achieves 0.255 accuracy and 0.151 macro-F1. These results are above random guessing but remain substantially below the Top-N=15 RemoteCLIP ensemble, confirming that simple RGB thresholds cannot capture the fine-grained visual distinctions among deep water, turbid water, rocky substrate, emergent vegetation, and submerged aquatic vegetation.

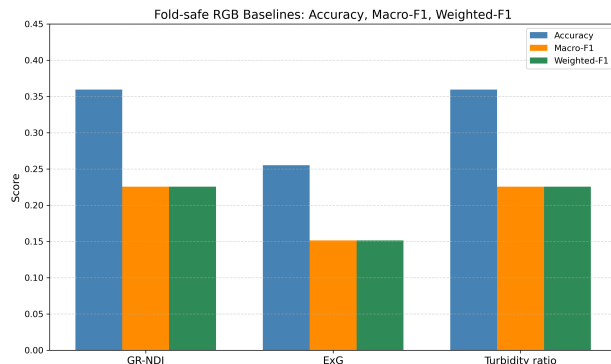


Figure 4: Fold-safe RGB baseline comparison. GR-NDI and the turbidity ratio perform similarly, while ExG performs worst. All RGB-only proxies remain far below the Top-N=15 RemoteCLIP ensemble, illustrating the limitations of scalar RGB heuristics for fine-grained hydrological classification.

Fold-wise Confidence Intervals

For completeness, we also computed fold-wise confidence intervals for accuracy and macro-F1. These intervals were narrow across all Top-N values, indi-

cating stable performance across folds.

- Accuracy CI (Top-N=15): 0.691 ± 0.012
- Macro-F1 CI (Top-N=15): 0.683 ± 0.015

Fold-wise means differ slightly from the pooled values in Table 1 because macro-F1 is computed independently within each fold before averaging.

Conclusion

The supplementary analyses confirm that prompt ensembling improves classification performance and Brier-score calibration relative to single-prompt inference. ECE improves most strongly at intermediate ensemble sizes and remains below the Top-1 setting for larger ensembles. The RGB-only baselines perform above chance in accuracy but remain far below RemoteCLIP, especially in macro-F1, supporting the need for semantic vision-language representations for fine-grained aquatic feature classification.